

MAT 426, Class 17 Homework: Orthogonal Diagonalization
(Due Tuesday, April 14, 2026)

1. Orthogonally diagonalize the matrix A by finding an orthogonal matrix Q and a diagonal matrix D such that $Q^T A Q = D$.

$$(a) \ A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad (b) \ A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix} \quad (c) \ A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad (d) \ A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

2. If A is an invertible matrix that is orthogonally diagonalizable, show that A^{-1} is orthogonally diagonalizable.
3. If A and B are orthogonally diagonalizable and $AB = BA$, show that AB is orthogonally diagonalizable.
4. Suppose $A = P R P^{-1}$ where P is orthogonal and R is upper triangular. Show that if A is symmetric, then R is symmetric and hence is a diagonal matrix.